19 EXAM PRACTICE

1 After the reduction of nitrobenzene to phenylamine, using tin and hydrochloric acid, an excess of sodium hydroxide is added.

What is the purpose of the sodium hydroxide?

A to dry the product

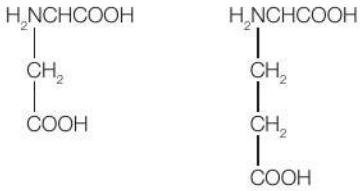
B to liberate phenylamine

C to lower the boiling point for subsequent distillation

D to precipitate tin(II) hydroxide [1]

(Total for Question 1 = 1 mark)

2 The amino acids aspartic acid and glutamic acid can react with each other to form amide linkages.



aspartic acid glutamic acid

What is the maximum number of different compounds, each containing only one amide linkage, that can be formed from one molecule of aspartic acid and one molecule of glutamic acid?

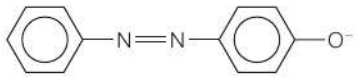
A 1 **B** 2 **C** 4 **D** 6 [1]

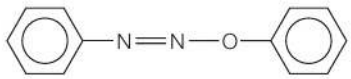
(Total for Question 2 = 1 mark)

3 What is the formula of the organic product obtained from the coupling reaction of the benzenediazonium ion, $C_6H_5N_2^+$, and the phenoxide ion, $C_6H_5O^-$ in alkaline solution?

A 

B 

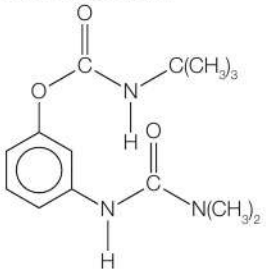
C 

D 

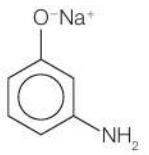
[1]

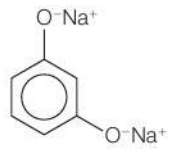
(Total for Question 3 = 1 mark)

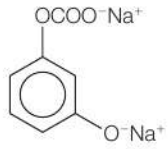
4 The structure of a herbicide called karbutilate may be represented as:

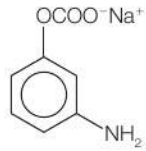


Which compound would be formed by prolonged boiling of karbutilate with aqueous sodium hydroxide?

A 

B 

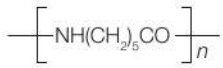
C 

D 

[1]

(Total for Question 4 = 1 mark)

5 Nylon 6 has the following formula:



What is the product of the acidic hydrolysis of nylon 6?

A $H_2N(CH_2)_5COOH$

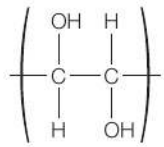
B $H_2N(CH_2)_5COO^-$

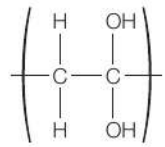
C $H_3N^+(CH_2)_5COOH$

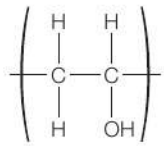
D $H_3N^+(CH_2)_5COO^-$ [1]

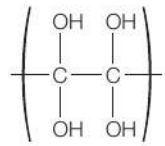
(Total for Question 5 = 1 mark)

6 The addition polymer poly(ethenol) is water-soluble. Which is the repeating unit of poly(ethenol)?

A 

C 

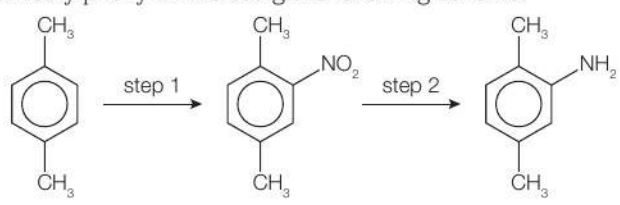
B 

D 

[1]

(Total for Question 6 = 1 mark)

7 A student proposed to make the compound 2,5-dimethylphenylamine using the following scheme:



1,4-dimethylbenzene 2,5-dimethylphenylamine

(a) (i) Name the two reagents needed for Step 1. [2]

(ii) Suggest why 1,4-dimethylbenzene is more reactive than benzene in reactions such as Step 1. [2]

(b) What type of reaction occurs in Step 2? [1]

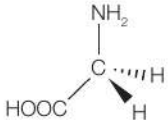
(c) 2,5-Dimethylphenylamine can be used to make azo-dyes.

State the reagents and conditions needed to make an azo-dye from 2,5-dimethylphenylamine and phenol.

Include equations for the organic reactions. [5]

(Total for Question 7= 10 marks)

8 The simplest amino acid is glycine.



glycine

(a) Give the systematic (IUPAC) name for glycine. [1]

(b) (i) Give the structural formula for the zwitterion of glycine in the solid state. [1]

(ii) Explain why glycine has a relatively high melting temperature for a molecule with such a low molar mass. [2]

(c) Alanine, $H_2NCH(CH_3)COOH$, is another amino acid.

Alanine has optical isomers (enantiomers).

(i) Draw diagrams to show the two optical isomers of alanine. [2]

(ii) Explain how the two optical isomers can be distinguished from each another. [2]

(d) Alanine and glycine react together in a condensation reaction to form two dipeptides.

(i) State why this is described as a condensation reaction. [1]

(ii) Give the structural formula for each of the two dipeptides that can be formed between alanine and glycine. [2]

(Total for Question 8 = 11 marks)

9 Butylamine, $CH_3CH_2CH_2CH_2NH_2$, and phenylamine, $C_6H_5NH_2$, both behave as bases.

(a) (i) State the feature of an amine molecule that allows it to act as a base. [1]

(ii) Explain why phenylamine is a weaker base than butylamine. [2]

(iii) Give the formulae of the salts formed when butylamine reacts with sulfuric acid and with ethanoic acid. [2]

(b) Phenylamine is formed by the reduction of nitrobenzene, $C_6H_5NO_2$.

State the reagents used for this reaction. [2]

(c) When phenylamine is reacted with a mixture of sodium nitrite, $NaNO_2$, and concentrated hydrochloric acid at a temperature between 0 and 5 °C, a diazonium ion is formed.

If phenylamine is added to the cold solution containing the benzenediazonium ion, a coupling reaction takes place similar to that between the benzenediazonium ion and phenol.

(i) Draw the structure of the diazonium ion, clearly displaying the functional group present in the ion. [1]

(ii) Suggest a structure for the organic product of the coupling reaction between the benzenediazonium ion and phenylamine. [1]

(iii) Suggest a use for this organic product. [1]

(Total for Question 9 = 10 marks)

10 Naturally occurring amino acids have the formula $H_2N(RCH)COOH$.

(a) In the amino acid glycine, the formula of the R group is —H.

Write the structural formula of the zwitterion ion formed by glycine. [1]

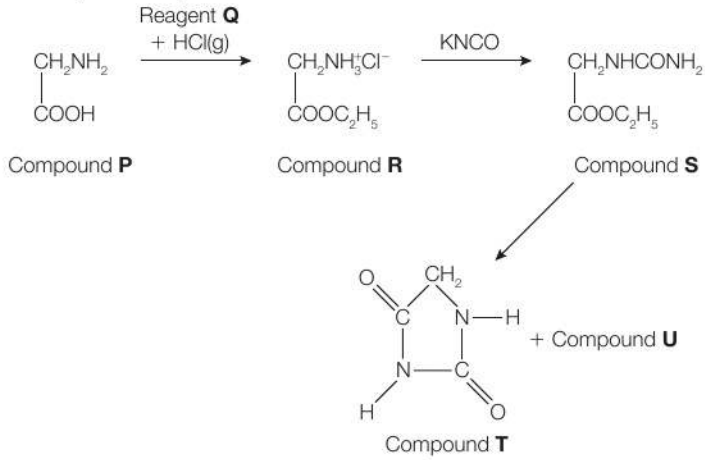
(b) In the amino acid serine, the formula of the R group is $—CH_2OH$.

Write the structural formula of serine at pH 1.0 and pH 10.0. [2]

(c) In the amino acid alanine, the formula of the R group is $—CH_3$.

Draw a section of the condensation polymer that can be formed from alanine. Show two repeat units of the polymer and display the link between them. [2]

(d) Compound **P** can be converted in three steps into a cyclic compound **T**.



(i) Give the systematic name for compound **P**. [1]

(ii) Name reagent **Q**. [1]

(iii) Name the two types of reactions that are occurring when compound **P** is converted into compound **S**. [2]

(iv) Suggest the identity of compound **U**. [1]

(v) When compound **T** reacts with hot dilute hydrochloric acid, compound **P** is formed, as well as two other products.

Suggest the type of reaction that is occurring. [1]

(vi) Suggest the identity of the two products that also form in the reaction in (d)(v). [2]

(Total for Question 10 = 13 marks)